

The CI platform is structured as a 7-layer intelligence framework that transforms raw call transcripts into evidence-backed operational strategies through AI analytics, behavioral modeling, and continuous learning.

The platform operates as a closed-loop intelligence system:

Call Data -> Pattern Discovery -> Strategy Creation -> Strategy Tracking -> Continuous Recalibration

1. Pre-Call Pattern Layer (Dynamic IVR intelligence) (Implemented at the end)
 - a. Use historical CI insights to inform dynamic IVR intelligence, not live agent scripting.
 - b. Inputs:
 - i. Known recurring issues
 - ii. Call intent distributions
 - iii. Successful resolution paths
 - c. Outputs:
 - i. Smarter IVR routing
 - ii. Issue-aware call tagging
 - iii. Early metadata enrichment
 - d. This layer is downstream of CI learning, not a prerequisite.
2. Transcript Processing Layer (Fact Gathering Engine) 6 modular components:
 - a. SPS-provided call summaries:
 - i. Why did the customer call?
 - ii. What did the agent do?
 - iii. What did the agent ask the customer to do?
 - iv. What was left unaddressed?
 - v. Sentiment over time:
 1. Opening sentiment
 2. Mid-call sentiment
 3. Closing sentiment
 4. Sentiment shift(screen based on this)
 - b. Transcript Ingestion
 - i. Pull transcript
 - ii. Mask sensitive info
 - iii. Attach metadata
 - iv. Store raw version
 - c. Call Segmentation Engine
 - i. A call can contain multiple goals/issues.
 - ii. Segment transcript into chunks:
 - iii. Example:

- iv. Chunk 1 - Payment Issue
 - Chunk 2 - Account Unlock
 - Chunk 3 - Late Fee Dispute
 - v. Each segment gets its own:
 - 1. Topic
 - 2. Sentiment trajectory
 - 3. resolution status
 - 4. agent behavior
 - 5. Unaddressed items
- 3. Layer 3: Behavioral & Outcome Extraction
 - a. Component 1: Why Did the Customer Call?
 - i. Techniques:
 - ii. Topic modeling
 - iii. Embedding clustering
 - iv. Supervised classification (if labels exist)
 - v. Hybrid: known patterns DB + unsupervised discovery
 - vi. Outputs:
 - 1. Canonical issue taxonomy
 - 2. Confidence scores
 - 3. Issue recurrence tracking
 - b. Component 2: What did the agent do? ADD: Behavioral and topic-wise
 - i. Extract behavioral signals:
 - 1. Examples:
 - a. Explained the timeline early
 - b. Interrupted customer
 - c. Escalated quickly
 - d. Offered an empathy statement
 - e. Provided clear next steps
 - f. Asked verification questions
 - g. Placed the caller on hold
 - ii. Behaviors are:
 - 1. Time-stamped
 - 2. Issue-linked
 - 3. Comparable across agents
 - c. Component 3: What Did Agent Ask Caller To Do?
 - i. Identifying these will help with:
 - 1. Resolution success
 - 2. Repeat calls
 - 3. Customer sentiment
 - d. Component 4: What Was Left Unaddressed? (Repeat Call Prediction)
 - e. Component 5: Sentiment Over Time
 - i. Not just one sentiment score.
 - ii. Track:
 - 1. Opening sentiment

- 2. Mid-call sentiment
 - 3. Closing sentiment
 - 4. Sentiment shift
- f. Component 6: Call outcome Classification
 - i. For each segment have:
 - 1. Positive/Neutral/Negative
 - 2. Resolved v/s Unresolved
 - 3. Escalated v/s contained
 - 4. Repeat-call likelihood
- 4. Analytical Intelligence Layer (Pattern + Correlation Engine)
 - a. The Triple Engine (Problem, Agent Behavior, Outcome) e.g. (Payment Confusion, Explained Timeline Early, Positive Outcome)
 - i. Aggregate across thousands of calls.
 - ii. Then compute:
 - 1. Correlation strength
 - 2. Lift
 - 3. Statistical significance
 - 4. Before/after comparison
 - b. Call Pattern Graph (Knowledge Graph):
 - i. visualizes relationships between issues, behaviors, and outcomes as a network graph.
 - ii. Edges become thicker as correlation strength increases.
 - iii. This produces a knowledge graph of call center behavior, allowing managers to visually explore how actions influence outcomes.
 - c. Recurrence & Trend Detection:
 - i. Emerging issue clusters
 - ii. Volume spikes
 - iii. New pattern detection
 - iv. Declining issues
 - v. Strategy performance drift
 - d. Alerting System
 - i. generates alerts when anomalies occur.
 - ii. Escalations increased 22% today
 - iii. New issue cluster detected
 - iv. Repeat call risk rising for billing issues
- 5. Evidence Pack (needed by Anthony as described in Description):
 - a. Contains:
 - i. Time window
 - ii. Filters applied
 - iii. Sample Size
 - iv. Counts and trends
 - v. Representative Call IDs
 - vi. Metrics Snapshot
 - vii. Derived Signals

6. Strategy & Continuous Learning Layer

- a. Insights discovered by the analytical layer are converted into operational strategies.
- b. Example Strategy : Explain payment timelines within the first 30 seconds.
- c. Strategy Lifecycle:
 - i. Proposed
 - ii. Accepted
 - iii. Implemented
 - iv. Evaluating
 - v. Closed

7. Dashboards

- a. Quick overview
 - i. Only have a few performance metrics
 - ii. Links to indepth versions for each metric
 - iii. List of common problems
- b. Problem dashboard
 - i. List of common phrasings
 - 1. Performance by each phrasing
 - 2. Relative differences
 - 3. Link to transcripts that use each phrasing
 - ii. Problem frequency

8. Presentation & Governance Layer

- a. Should have all these screens (some taken from the description):
 - i. Executive Overview
 - ii. Daily Insights Pulse
 - iii. Monthly Recalibration
 - iv. Issue Explorer
 - v. Call Drilldown
 - vi. Strategy Management
 - vii. Strategy Effectiveness
 - viii. Reports & Export
 - ix. Sentiment Journey Map
 - x. Admin & AI Governance
 - xi. Voice to Insight
 - xii. Natural Language Ask CI (Like a Chatbot)
 - 1. Managers can ask questions in natural language.
 - 2. Examples:
 - 3. Why are escalations increasing?
 - 4. What agent behaviors reduce repeat calls?
 - 5. CI responds using RAG (Retrieval Augmented Generation) on analytics tables and evidence packs.